MIL-T-13057C
4 August 1981
SUPERSEDING
MIL-T-13057B
2 June 1976

THERMOMETERS, SELF-INDICATING, LIQUID-IN-GLASS

(FOR PRECISION LEVEL ROD)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two sizes of etched-stem, liquid-in-glass thermometers graduated in Celsius (C) degrees.

1.2 Classification. The thermometers shall be the following sizes, as specified (see 6.2):

Size I - 0.312-inch offset
Size II - 0.75-inch offset

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FSC 6685

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation, Methods of.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The thermometers shall be etched-stem, liquid-in-glass, total-immersion type, graduated in Celsius degrees, as shown in Figure 1, and as specified herein.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.3).

3.3 Storage. The thermometer shall not be damaged by storage in any ambient temperature from minus 40° F to plus 160° F (minus 40° C to plus 71° C).

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3.4 Thermometer tube.

3.4.1 Stem. The stem shall be magnifying-lens type glass with a ring top and a strip of white glass at the back to increase reflection of light. The stem shall be designed to increase the angle of vision and effective width of the thermosensitive fluid column and to accommodate the graduations and numbers on the front of the tube. The stem shall be annealed.

3.4.2 Bulb. The bulb shall be normalized glass, cylindrical in shape, or blown from the stem.

3.4.3 Thermosensitive fluid. The bulb and tube shall contain a sufficient quantity of fluid so that, when correctly calibrated, the height of the fluid in the tube shall indicate the temperature of the medium surrounding the thermometer within the range and accuracy specified herein. The type of fluid used shall minimize any tendency for the liquid column to separate. A non-fading dye may be used to color the fluid to allow a contrast for reading.

3.4.4 Characteristics. The thermometer tube shall conform to the following characteristics:

Immersion	- Total
Temperature range	- Minus 38° C to plus 52° C
Graduations	- 1° C
Width of graduation lines	- 0.004, plus 0.001 minus 0.000 inch
Longer graduation lines at each	- 5° C
Graduations numbered at each multiple of	- 10° C
Scale error at any point when standardized shall be not more than	- 1° C
Standardization	- Ice point and every 20° C
Special inscription on thermometer	- Letter "C" etched above the upper limit of the scale.

3.4.5 Graduations. The graduation marks and figures shall be etched on the stem, filled with an indelible pigment of distinctive color, and shall be of sufficient size to be clearly legible from a distance of 18 inches. For temperatures of zero degrees and above, the color shall be black. For temperatures below zero, the color shall be red. The pigment shall not be affected by submersion for 10 days in a solution of not less than 75 percent sulfuric acid or a commercial grade of toluene at 100° F (38° C).

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3.5 Identification. Each thermometer shall be identified in accordance with MIL-STD-130.

3.6 Size I thermometers. Size I thermometers shall have a 0.312-inch offset as shown in Figure 1.

3.7 Size II thermometers. Size II thermometers shall have a 0.75-inch offset as shown in Figure 1.

3.8 Workmanship. The thermometers and thermosensitive fluid shall be free from gas bubbles and foreign material. The thermometers shall be clean and free from pits, chips, sharp edges, or other defects which would prove detrimental to accuracy and appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined for the defects specified in 4.4.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for the minor defect.

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4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2. AQL shall be 2.5 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The thermometers shall be examined as specified herein for the following defects:

Major

101. Material not as specified.
102. Stem not as specified.
103. Separation or bubbles in the thermosensitive fluid.
104. Temperature range not as specified.
105. Graduations not as specified.
106. Workmanship not as specified.

Minor

201. Identification marking on the thermometer missing, incomplete, or illegible.

4.4.2 Tests.

4.4.2.1 Test conditions. A thermometer which has been certified by the National Bureau of Standards to have an accuracy of plus or minus 0.1° C shall be used in the test specified in 4.4.2.3.

4.4.2.2 Storage. Sample thermometers shall be placed in a cold chamber at a temperature of minus 40° F (minus 40° C) for not less than 3 days without solar radiation. The thermometers shall then be removed, allowed to return to room temperature, and placed in an oven at plus 160° F. (plus 71° C) for 4 hours. The thermometers shall be removed from the oven, allowed to return to room temperature, and examined. Any evidence of breaks, cracks, or separation bubbles in the thermosensitive fluid shall constitute failure of this test.

4.4.2.3 Standardization. The certified thermometer and the sample thermometers shall be totally immersed in a well-stirred water bath and compared at ice point (0° C) and at every 20° C interval. Failure of the sample thermometers to indicate the same reading as the certified thermometer within 1° C shall constitute failure of this test.

4.4.2.4 Marking. The thermometers shall be submerged in either a solution of not less than 75 percent sulfuric acid or a commercial grade of toluene at plus 100° F (plus 38° C) for a period of 10 days. At the end of the test period, the thermometers shall be examined. Any damage to or loss of pigmentation reducing legibility of graduations or numbers shall constitute failure of this test.

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4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107.	Methods or containers not as specified. Each incorrect method or container shall constitute one defect.
108.	Quantities placed in consolidated container not as specified for Level A.
109.	Thermometers of unlike size consolidated together for Level A.
110.	Fiberboard boxes not closed as specified for Level B.
111.	Commercial packaging not in accordance with referenced documents.
112.	Marking illegible, incorrect, incomplete, or missing.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each thermometer shall be preserved in accordance with MIL-P-116, Method III.

5.1.1.1 Consolidation. Thermometers of like size, shall be consolidated together in a close fitting box conforming to PPP-B-636, W6c, style optional, in quantities as specified (see 6.2). Box closure shall be in accordance with the appendix to the box specification.

5.1.2 Commercial. Commercial preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. The thermometers, preserved as specified in 5.1, shall be packed in close fitting boxes conforming to PPP-B-601, Overseas Type, style optional. Box closure and strapping shall be in accordance with the appendix to the box specification.

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5.2.2 Level B. The thermometers preserved as specified in 5.1, shall be packed in close fitting boxes conforming to PPP-B-636, V3c, style optional. The boxes shall be closed in accordance with Method V of the appendix to the box specification.

5.2.3 Commercial. The thermometers shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for Commercial protection shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The thermometers are intended for use with precise geodetic level rods conforming to MIL-R-10916. The Size I thermometers are used with rods which have a 1.125 inch thickness and a 1.250 inch recessed invar strip. The Size II thermometers are used with rods which have a 1.094 inch thickness and an invar strip flush with the surface.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of thermometer required (see 1.2).
(c) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(d) Quantities to be consolidated together (see 5.1.1.1).

6.3 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

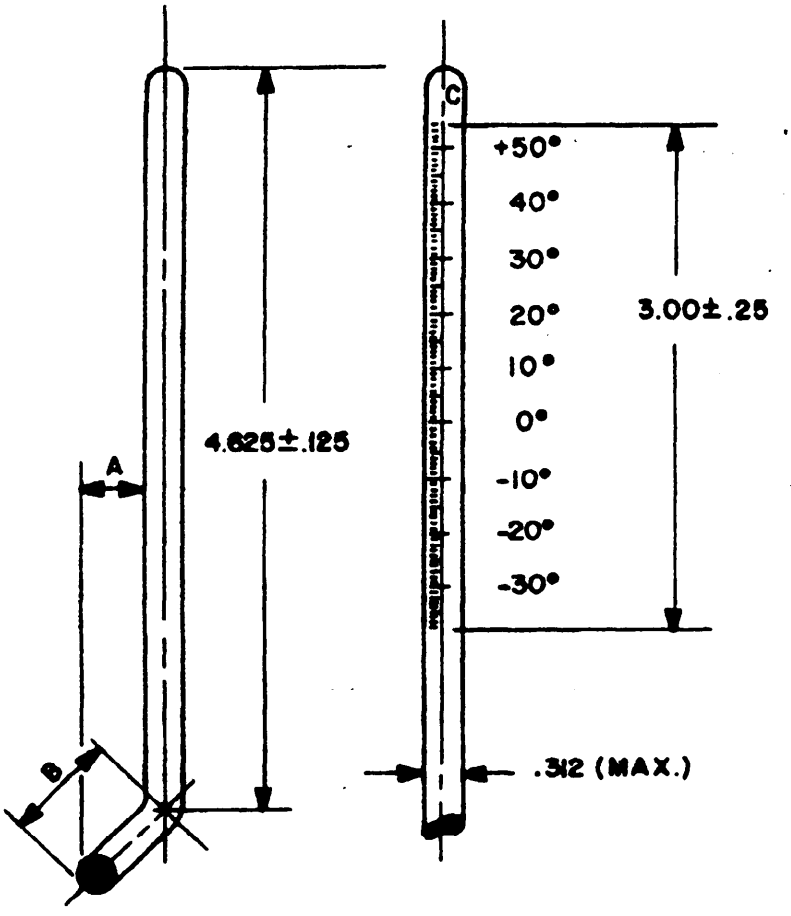
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NOTE: ALL DIMENSIONS ARE IN INCHES.

	A	B	DEVELOPED LENGTH (APPROX)
SIZE I	.312 ±.031	.625 ±.031	5.250
SIZE II	.750 $^{+.000}_{-.125}$	1.250 $^{+.000}_{-.125}$	5.875

FIGURE 1. Thermometer, self-indicating, liquid-in-glass (for precision level rod).

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DOCUMENT IDENTIFIER (Number) AND TITLE MIL-T-13057C Thermometers, Self-Indicating, Liquid-In-Glass (For Precision Level Rod)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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